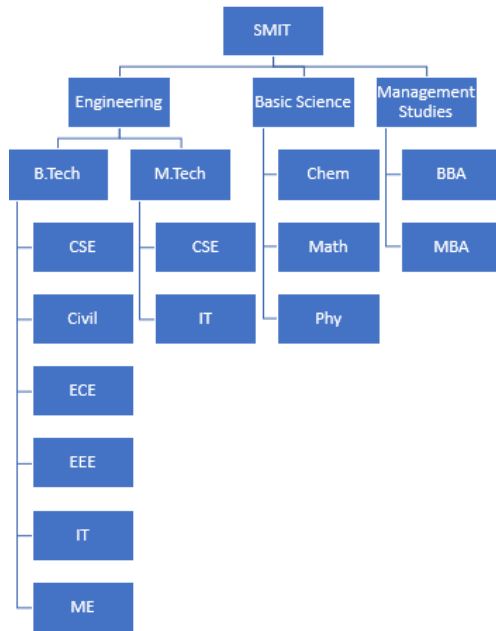


The questions in *Italic* are for practice purpose, no need to solve those questions, unless stated by your subject teacher specifically.

Q1. Create following directory hierarchy using Linux shell commands.



Q2. Write programs to display message in following manner using `\t` and `\n` if applicable.

- a. HelloWorld
- b. Hello World
- c. Hello
World

Q3. Write a program to demonstrate method for declaring and initializing variables in C.

Q4. Write a program in C to demonstrate working of `printf( )` and `scanf( )` in

Q5. Write an algorithm to find sum and average of five numbers.

Q6. Write an algorithm to find the largest among two different numbers entered by user.

Q7. Draw a flowchart to find sum and average of five numbers.

Q8. Write an algorithm to check if a given number is odd or even. Draw a flowchart to demonstrate the same.

Q9. Write a program to demonstrate the use of `float`, `char`, `double` & `unsigned int`.

Q10. Write a program to demonstrate the use of `sizeof( )`, `const`, `#define`.

Q11. Write a program in C to demonstrate the use of basic arithmetic operators +, -, *, /, %. Write an algorithm and draw the flowchart for the same.

Q12. Write a program to compute Simple Interest. Write an algorithm & draw the flowchart for the same.

Q13. Write a program to compute Compound Interest. Write an algorithm & draw the flowchart for the same.

HINT: Use float or double data type.

Q14. Write a program to swap two variables using a third variable. Write an algorithm & draw the flowchart for the same.

Q15. Write a program to swap two variables without using a third variable. Write an algorithm & draw the flowchart for the same.

Q16. Write a program in C to convert temperature value from Celsius to Fahrenheit and vice versa.

Q17. Write a program in C to take time value in second's format as user input and display its equivalent in the following format.

hours: minutes :seconds

Example:

Input: 3666

Output : 1hr : 1min : 6sec

Q18. Write a program in C to calculate and display the Net Salary (NS) of an employee. The Dearness Allowance (DA) is 40% of Basic Salary (BS), House Rent Allowance (HRA) is 20% of basic. Tax deduction is 10 % of gross salary. Calculate the net salary using the formula give below.

Gross Salary (GS) = DA + HRA + BS

Net Salary = GS – Tax amount

Q19. Write a program in C to find the greatest among three numbers using ternary operator. Write an algorithm & draw the flowchart for the same.

Q20. Write a program in C to check whether an alphabet is vowel or consonant using ternary operator. Write an algorithm & draw the flowchart for the same.

Q21. Write a program in C to find the greatest among three numbers using if-else construct.

Q22. Write a program in C to check whether an alphabet is vowel or consonant using if-else construct.

Q23. Write a program in C to find the roots of a quadratic equation using if-else construct.

Q24. Write a program in C to read marks for physics, chemistry and biology, calculate the average marks in science subject and display the grade in science using the table given below (using switch- case).

Marks in Science(out of 100)	Grade
90 and above	S
Below 90, 80 and above	A
Below 80, 70 and above	B
Below 70, 60 and above	C
Below 60 , 50 and above	D
Below 50 , 40 and above	E
Below 40	F

Q25. Write a program in C to design a basic calculator with addition, subtraction,multiplication, power and division facility. Implement using switch case

Q26. Write a program in C to read marks for physics, chemistry and biology, calculate the average marks in science subject and display the grade in science using the table given below using else-if ladder.

Marks in Science(out of 100)	Grade
90 and above	S
Below 90, 80 and above	A
Below 80, 70 and above	B
Below 70, 60 and above	C
Below 60 , 50 and above	D
Below 50 , 40 and above	E
Below 40	F

Q27. Write a program in C to design a basic calculator with addition, subtraction, multiplication, power and division facility. Implement using else-if ladder.

Q28. Write a program in C to calculate the sum of N natural numbers. $1+2+3+\dots+N$

Q29. Write a program in C to calculate the sum of series given below. $1^2+2^2+3^2+\dots+N^2$

Q30. Write a program in C to calculate the sum of series given below. $11+22+33+\dots+NN$

Q 31. Write a program in C to find the sum of digits in a number.

Q 32. Write a program in C to find the reverse of a number and check whether the number is palindrome or not.

Q33. Write a program in C to convert decimal number to its binary equivalent.

Q 34. Write a program in C to display following patterns using loop.

```
*  
**  
***  
****
```

35 Write a program in C to display given pattern using loop.

```
  *  
 **  
 ***  
 ****  
 *****
```

Q36 Write a program in C to display given pattern using loop.

```
  *  
 **  
 ***  
 ****  
 *****
```

Q37 Write a program in C to display given pattern using loop.

```
1111  
 222  
  33  
   4
```

Q38 Write a program in C to display given pattern using loop.

```
1111  
 222  
  33  
   4
```

Q39. Write a program in C to count number of digits in a number using loop.

Q40. Write a program in C to count number of students who have scored Grade S, A, B, C, D,E & F.

Q41. Write a program in C to condense a number into single digit.

Example:

Input : 12345

$4+5+6+7+8+9 = 39$

$3+9 = 12$

$1+2 = 3$

Output = 3

Q42. Write a program in C to find factorial of a number using loop

Q43. Write a program in C to find Fibonacci series till N terms

Q44. Write a program in C to find the sum and average of elements in 1D float array.

Q45. Write a program in C to find the smallest and largest number in 1D double

Q46. Write a program in C to reverse elements of 1D array and store in another array.

Q47. Write a program in C to demonstrate Linear search in 1D array.

Q48. Write a program in C to search an element in 1D array using Binary search technique.

Q49. Write a program in C to sort all integer elements of 1D array using Bubble Sort technique.

Q50. Write a program in C to reverse elements of an array and store it in same array.

Q51. Write program in C to display all the odd elements in 1D integer array.

Q52. Write a program in C to display all the elements present in even location in 1D array.

Q53. Write a program in C to find the sum of all odd elements in 1D array.

Q54. Write a program in C to count number of positive, negative and zero elements in 1D array.

Q55. Write a program in C to display all the prime numbers present in 1D array.

Q56. Demonstrate the use of puts(), putchar(), gets(), getchar().

Q57. Write a menu driven program in C to perform following basic operations on a string without using built-in function available under <string.h>.

Q58. Write a menu driven program in C to perform following basic operations on a string using built-in function available under <string.h>.

Q59. Write a program to check whether a string is a palindrome or not.

Q60. Write a program in C to compare two strings without using <string.h> functions.

Q61. Write a program in C to check if a substring is present in a string without using <string.h> functions.

Q62. Write a program in C to find length of a string using function available in <string.h>

Q63. Write a program in C to reverse a string & check if it is a palindrome using function available in <string.h>

Q64. Write a program in C to copy string from one memory to another with or without using function available in <string.h>.

Q65. Write a program in C to find sum of elements of two 2D array. The sum must be stored in a new 2D array.

Q66. Write a program in C to find difference of elements of two 2D array. The difference must be stored in a new 2D array.

Q67. Write a program in C to find product of elements of two 2D array. The product must be stored in a new 2D array.

Q68. Write a program in C to transpose a 2D array and display it.

Q69. Write a program in C to display the elements of left diagonal in a 2D

Q70. Write a program in C to find maximum & minimum element in a 2D array.

Q71. Write a program in C to display elements of 2D array in following format:-

a. Row major

b. Column major

Q72. Write a program in C to display sum of all the right diagonal elements of a 2D

array. Q73. Write a program in C to replace all the left diagonal elements with zero.

Q74. Write a program in C to display all the elements in the triangle above the left diagonal.

Q75. Write a program in C to display all the elements in the triangle below the left

diagonal. Q76. Write a program in C to replace all the elements in even rows with zero.

Q77. Write a program in C to display all the odd elements present in a 2D array.

Q78. Write a program in C to count all the even elements present in a 2D array.

Q79. Write a program in C to add two numbers using a function.

Q80. Write a program in C to simulate a digital calculator using parameterized function for every operation & along with return values.

Q81. Write a program in C to display the Fibonacci series upto nth terms using parameterized function.

Q82. Write a program in C to find factorial of a number using recursion.

Q83. Write a program in C to create a function named c_power() with parameters base and exponent to calculate base raised to the power of exponent. Do not use pow() function.

Q84. Write a program in C to create a parameterized function name c_prime() to find all the prime numbers

between a given range.

Q85. Write a program in C to create a parameterized function to convert decimal to binary. The binary equivalent must be displayed from main().

Q86. Write a program in C to create a parameterized recursive function to find sum on N natural numbers.

Q87. Write a program in C to reverse a number using function recursion.

Q88. Write a program in C to create a parameterized function and check if a given number is Armstrong number.

Q89. Find sum of elements in an array using user defined functions by passing array in a function.

Q90. Write a program in C to swap two numbers using call by reference.

Q91. Write a program in C to perform bubble sort of elements in a 1D array using pointers.

Q92. Write a program in C to search element in 1D array using linear searching techniques.

Implement all the operations on the array using pointer.

Q93. Write a program in C to reverse a string in 1D array using pointers.

Q94. Write a program in C to Display elements of 2D array using pointers.

Q95. Write a program in C to calculate sum & average of two numbers using pointers.

Q96. Write a program in C to count number of vowels and consonants in a string using pointers.

Q97. Write a program in C to read double elements in C and display elements along with their index and address using pointers and array.

Q98. Write a program in C to find area of a circle using pointers.

Q99. Write a program in C to find lowest common multiple in C using pointers and function.

Q100. Write a program in C to swap all the elements of two 1d array and display them using functions and pointers.

Q101. Write a program in C to create a structure named MOVIE to store following details of one movie and display it:-

- Movie title
- Movie producer
- Budget
- Duration of movie in hours

Q102. Write a program in C to create a structure named MOVIE to store following movie details and display it. The structure must be able to store movie details for 100 movies.

- Movie title
- Movie producer
- Budget
- Duration of movie in hours

Q103. Write a program in C to create a structure named LAPTOP_DETAILS to store following configuration details of a laptop.

- Brand
- Model

- RAM size
- Hard disk size
- Color
- Screen size

Create a function `accept( )` to store details for fifty laptops in the structure. Create another function `display( )` to display the laptop configuration details for all the laptops in a tabular format.

Q104. Write a program in C to open a file named `my_c_file` and store a string "SIKKIM MANIPAL INSTITUTE OF TECHNOLOGY" .

Q105. Reopen the file `my_C_file`, read & display its content.

Q106. Write a program in C to create a file named "student_details", store name and registration number of ten students. Reopen the file and display the details in a tabular format.